



INVERTEBRATES

SPONGES

PHYLUM Porifera CLASSES 3 ORDERS 24 FAMILIES 127 SPECIES C.10,000

CNIDARIANS

PHYLUM Cnidaria CLASSES 6 ORDERS 24 FAMILIES 300 SPECIES C.11,000

FLATWORMS

PHYLUM Platyhelminthes CLASSES 6 ORDERS 41 FAMILIES 424 SPECIES C.30,000

SEGMENTED WORMS

PHYLUM Annelida CLASSES 4 ORDERS 17 FAMILIES 130 SPECIES C.18,000

ROUNDWORMS

PHYLUM Nematoda CLASSES 2 ORDERS 17 FAMILIES 160 SPECIES C.26,000

MINOR PHYLA

ROTIFER



Invertebrates are classified in about 30 phyla, which vary considerably in size. In this book, major phyla are treated separately, but a few

minor phyla also appear on pages 544–5. Almost all these minor phyla contain marine animals or ones that live in damp habitats.

COMB JELLIES

PHYLUM Ctenophora SPECIES C.200

PEANUT WORMS

PHYLUM Sipuncula SPECIES C.150

ROTIFERS

PHYLUM Rotifera SPECIES C.2,000

BRYOZOANS

PHYLUM Bryozoa SPECIES C.6,000

RIBBON WORMS

PHYLUM Nemertea SPECIES C.1,400

BRACHIOPODS

PHYLUM Brachiopoda SPECIES C.400

ARROW WORMS

PHYLUM Chaetognatha SPECIES C.150

WATER BEARS

PHYLUM Tardigrada SPECIES C.1,000

SPOONWORMS

PHYLUM Echiura SPECIES C.200

HORSESHOE WORMS

PHYLUM Phoronida SPECIES C.20

VELVET WORMS

PHYLUM Onychophora SPECIES C.180

HEMICHORDATES

PHYLUM Hemichordata SPECIES C.130

10 OTHER MINOR INVERTEBRATE GROUPS

MOLLUSCS

PHYLUM Mollusca CLASSES 7 ORDERS 53 FAMILIES 609 SPECIES C.110,000

ECHINODERMS

PHYLUM Echinodermata CLASSES 5 ORDERS 38 FAMILIES 173 SPECIES C.7,000

INVERTEBRATE CHORDATES

TUNICATE



Invertebrate chordates are animals that share some characteristics with vertebrates but lack a bony skeleton. There are 2 subphyla—tunicates (the majority of species), and lancelets. Tunicates have

a swimming tadpole stage but are baglike as adults; lancelets are mobile and in their internal anatomy bear strong resemblances to vertebrates.

TUNICATES

SUBPHYLUM Urochordata
CLASSES 3 ORDERS 7 FAMILIES 36 SPECIES C.2,900

LANCELETS

SUBPHYLUM Cephalochordata
CLASSES 1 ORDERS 1 FAMILIES 1 SPECIES 30

ARTHROPODS

PHYLUM Arthropoda

SPIDER-HUNTING WASP

Arthropods form the largest phylum in the animal kingdom. Insects make up the biggest subgroup, but the phylum

also contains 2 other giant classes, crustaceans and arachnids, which dwarf many phyla in the invertebrate world.

MANDIBULATES

SUBPHYLUM Mandibulata CLASSES 16 ORDERS 109 FAMILIES C.2,230 SPECIES C.1.2 million

HEXAPODS

SUPERCLASS Hexapoda CLASSES 4 ORDERS 32 FAMILIES C.1,047 SPECIES C.1.1 million

SPRINGTAILS

CLASS Collembola ORDERS 1 FAMILIES 32 SPECIES C.8,100

PROTURANS

CLASS Protura ORDERS 1 FAMILIES 7 SPECIES C.760

DIPLURANS

CLASS Diplura ORDERS 1 FAMILIES 8 SPECIES C.975

INSECTS

CLASS Insecta ORDERS 29 FAMILIES C.1,000 SPECIES C.1.1 million

BRISTLETAILS

ORDER Archaeognatha SPECIES C.470

SILVERFISH

ORDER Thysanura SPECIES C.570

MAYFLIES

ORDER Ephemeroptera SPECIES C.3,000

DAMSELFLIES AND DRAGONFLIES

ORDER Odonata SPECIES C.5,600

CRICKETS AND GRASSHOPPERS

ORDER Orthoptera SPECIES C.10,500

STONEFLIES

ORDER Plecoptera SPECIES C.3,000

ROCK CRAWLERS

ORDER Grylloblattodea SPECIES 30

EARWIGS

ORDER Dermaptera SPECIES C.1,900

STICK AND LEAF INSECTS

ORDER Phasmatodea SPECIES C.2,500

MANTIDS

ORDER Mantodea SPECIES C.2,300

COCKROACHES

ORDER Blattodea SPECIES C.4,600

TERMITES

ORDER Isoptera SPECIES C.3,000

WEB-SPINNERS

ORDER Embioptera SPECIES C.400

ANGEL INSECTS

ORDER Zoraptera SPECIES 43

BARKLICE AND BOOKLICE

ORDER Psocoptera SPECIES C.5,600

PARASITIC LICE

ORDER Phthiraptera SPECIES C.5,200

BUGS

ORDER Hemiptera SPECIES C.88,000

THRIPS

ORDER Thysanoptera SPECIES C.7,400

DOBSONFLIES AND ALDERFLIES

ORDER Megaloptera SPECIES C.300

SNAKEFLIES

ORDER Raphidioptera SPECIES C.200

ANTLIONS, LACEWINGS, AND RELATIVES

ORDER Neuroptera SPECIES C.11,000

BEETLES

ORDER Coleoptera SPECIES C.370,000

STREPSITERANS

ORDER Strepsiptera SPECIES C.580

SCORPIONFLIES

ORDER Mecoptera SPECIES C.550

FLEAS

ORDER Siphonaptera SPECIES C.2,500

FLIES

ORDER Diptera SPECIES C.150,000

CADDISFLIES

ORDER Trichoptera SPECIES C.10,000

MOTHS AND BUTTERFLIES

ORDER Lepidoptera SPECIES C.165,000

BEES, WASPS, ANTS, AND SAWFLIES

ORDER Hymenoptera SPECIES C.198,000

MYRIAPODS

SUPERCLASS Myriapoda CLASSES 2 ORDERS 21 FAMILIES 171 SPECIES C.13,150

CRUSTACEANS

SUPERCLASS Crustacea CLASSES 7 ORDERS 56 FAMILIES C.1,000 SPECIES C.70,000

CHELICERATES

SUBPHYLUM Chelicerata CLASSES 3 ORDERS 14 FAMILIES 675 SPECIES C.104,350

SEA SPIDERS

CLASS Pycnogonida ORDERS 1 FAMILIES 13 SPECIES C.1,330

HORSESHOE CRABS

CLASS Merostomata ORDERS 1 FAMILIES 1 SPECIES 4

ARACHNIDS

CLASS Arachnida ORDERS 12 FAMILIES 661 SPECIES C.103,000